

Ava Poland

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Skills

Mechanical Design & Cad: SolidWorks, Onshape, Fusion 360, Autodesk Inventor, Parametric Modeling, GD&T, ANSI Standards,

Manufacturing: FDM 3D Printing, CNC Machining, Laser Cutting, G-Code, DFM

Programming: Python, MATLAB, iLogic, Parametric CAD Automation

Experience

Mechanical Engineering Intern, Automatic Systems Incorporated – Kansas City, MO May 2026 – August 2026

- Automated 50+ conveyor model and drawing configurations in minutes by developing iLogic-driven Inventor templates, replacing hours-to-days of repetitive manual modeling and drafting.
- Designed and detailed Power & Free conveyor components including horizontal turns, vertical curves, return tracks, and specialized track assemblies for GM automotive manufacturing plants in Fairfax, KS and Arlington, TX.
- Modernized legacy conveyor engineering data by rebuilding decades-old AutoCAD drawings in Inventor and correcting dimensions, tolerances, obsolete components, constraints, and BOM information.
- Improved manufacturability and engineering data management by collaborating with shop personnel and using GD&T, DFM, Autodesk Vault, and Fusion Manage.

Engineering Graphics Teaching Assistant(ME 212), Kansas State University – Manhattan, KS Jan 2026 – Present

- Instructs students in mechanical drafting and CAD modeling through laboratory instruction covering ANSI engineering drawing standards and design communication.
- Demonstrates mechanical design techniques through hand-sketching and CAD-based modeling during instruction.
- Develops CAD-based examinations and assignments by modeling mechanical components that assess specific engineering graphics competencies.

Build Team Captain, FTC Tungsten Tigers – Blue Valley High School August 2022 – May 2024

- Co-led mechanical design of a multi-award-winning FTC robot by machining chassis and competition components.
- Developed competition mechanisms by designing a robotic claw and 3D-printing a paper-airplane launcher.
- Expanded team CAD and fabrication capabilities by mentoring members in CAD, prototyping, and robot construction.

Projects

Pi Car Pro - Custom Four-Finger Robotic Gripper 2026- Present

- Designing a custom robotic gripper and articulated arm in Solidworks for integration with a PiCarPro mobile platform.
- Developing a hybrid manufacturing strategy using TPU for compliant gripping surfaces and machined aluminum for higher-load structural components
- Programming servo motor control in Python to drive claw movement, turning response for precise grip actuation.

Egg Beater Combat Robot, Wildcat Combat Robotics – Manhattan, KS Aug 2026– Present

- Contributes to mechanical design of a competitive eggbeater-style combat robot by developing system layouts, component geometry, and iterative CAD designs.
- Supports robot subsystem integration and testing across mechanical and electrical systems, including drivetrain and other robot hardware.
- Creates adaptive component functionality by evaluating mechanical constraints and iterating designs for various environments.

Education

Kansas State University – B. S. in Mechanical Engineering, Minor in Physics

December 2028

SolidWorks Certified – C-R7Z4VQB269

May 2025